

Task 1

```
import os
import cv2
import matplotlib.pyplot as plt
from pathlib import Path
```

```
dataset_path = Path("BrainMRI/yes")

image = list(dataset_path.glob("*.jpg"))[0]
print("Image Path:", image)
```

Image Path: BrainMRI/yes/Y186.jpg

```
image = cv2.imread(image)

# Convert BGR to RGB
image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
```

```
_, segmented = cv2.threshold(
    gray,
    120,
    255,
    cv2.THRESH_BINARY
)
```

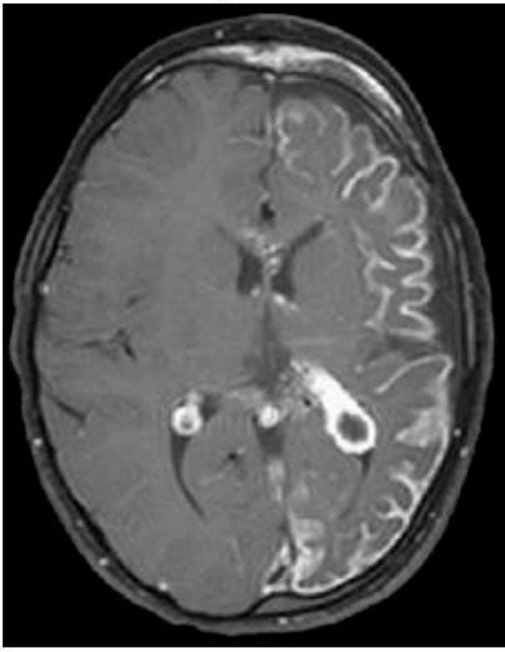
```
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(image_rgb)
plt.title("Original MRI")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(segmented, cmap="gray")
plt.title("Segmented MRI")
plt.axis("off")

plt.show()
```

Original MRI



Segmented MRI



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